Answer Key

MAHATMA GANDHI UNIVERSITY, KOTTAYAM
MGU-UGP (HONOURS) Regular EXAMINATION October 2025
Third SEMESTER
Discipline Specific Elective (DSE) - MG3DSEGEO200 - ENDOGENOUS EARTH PROCESSES
(2024 ADMISSION ONWARDS)

Part A
Short answer in 60 words.
Answer any 5 questions

1.State the age of the Earth. 4.6 billion years. Definition- 2 marks.

2.List the three major types of plate boundaries associated with tectonic activity - Convergent, Divergent and Transform plate boundary. Definition -2 marks.

3.Evaluate the size range of different pyroclastic materials-

Fine Ash-Less than 0.063 mm,Coarse Ash-0.063 mm up to 2 mm,Lapilli-2 mm to 64 mm,Bombs/Blocks-Greater than 64 mm -Definition -2 marks.

4.Mention which physiographic region of India is at the greatest risk of high-magnitude earthquakes.

Zone V-The Himalayan region of India is at the greatest risk of high-magnitude earthquakes-Parts of Jammu and Kashmir (Kashmir valley); Western part of Himachal Pradesh; Eastern part of Uttarakhand, Kutch in Gujarat; part of Northern Bihar; all northeastern states of India and the Andaman & Nicobar Islands. Definition and description -2 marks

5.Conservative plate boundaries where strike-slip or relative displacement of plates takes place across the spreading centres/ridges. -2 marks

6. Name one catastrophic event that could alter Earth's geology.
Volcanism, Meteorite impact, Earth quakes.- Definition -2 marks.

Part-B
Short notes in 250 words
Answer any 6 questions.
Each question carries 8 marks

7. Isostasy refers to the gravitational equilibrium between the Earth's lithosphere and the Asthenosphere. -1 mark

Post-glacial rebound or Glacial isostatic adjustment is a natural response of the Earth's crust to be uplifted as a result of the retreat of large ice sheets/glaciers. -1 mark

The lithosphere is assumed to float over the layer of asthenosphere, which is of a plastic nature.

Two classical models that describe isostasy are
Airy's model – variations in crustal thickness compensate for topographic differences.
Pratt's model – variations in crustal density compensate for elevation differences
-2 marks

The additional weight of huge ice sheets causes the lithosphere to sink deep into Asthenosphere.

When the weight of the ice sheets is removed as the ice melts down by climatic change of interglacial periods, the lithosphere rises or rebounds to regain equilibrium by the

redistri-bution of mantle material and asthenosphere.
Recent satellite data show measurable uplift in Antarctica and Greenland ice sheets thin due to climate change -2 marks

8. Evaluate the difference between pahoehoe and Aa lava flows.

Viscosity:*Pahoehoe* → Low viscosity lava; *Aa* → Higher viscosity lava.

Surface Appearance/Texture- *Pahoehoe* → Smooth, glassy, wrinkled, or "ropy" surface due to continuous flow as it cools. 2.5 marks

Aa → Jagged, rough, and rubbly surface formed when the solid crust breaks as molten lava moves beneath. 2.5 marks.

9. Explain the role of faults in the mechanism of earthquakes as described by the Elastic Rebound Theory.

Stress and Elastic Energy Storage (1 marks), Friction and Fault Locking (2 marks), Accumulation of Energy (2 marks),Rupture and Release – Elastic Rebound (2 marks),Conclusion – Role of Faults (1 mark).

10. Japan type orogeny is an island arc orogeny caused by the convergence of the oceanic plate (Eurasian and Pacific plates). 1.5 mark
One oceanic plate subducts beneath the other, and marked by trenches and an arcuate chain of volcanic islands called the Island Arc. -1.5 marks
Andes type orogeny is caused by the convergence of continental and oceanic plates (between the Pacific and South American plates). 1.5 mark
Oceanic plate subducts beneath the continental plate, forming deep trenches, and volcanic arcs along the continental margins -1.5 marks

11.Discuss geological similarities across continents as proof of continental drift.

Concept of Continental Drift (1 mark), Geological Evidence (3 marks)-Matching rock types and structures, Paleoclimatic Evidence (2 marks)-Glacial deposits of the same age found across continents now in tropical regions,Other climate indicators (desert sandstones, coral reefs, coal swamps) also align with expected climatic zones on Pangea. Fossil Evidence (2 marks)- Same plant and animal fossils (e.g., *Mesosaurus*, *Cynognathus*, *Lystrosaurus*, *Glossopteris*) found on widely separated continents.

12. Discussion on plate tectonism, and types of plate margins -3 marks
Seafloor spreading takes place along divergent margins where the mantle thermal upwelling takes place. - 2 marks
New crustal parts form along the spreading ridge, and that causes to push the oceanic plates in opposite directions across the axis of spreading. As a result the plates move over the face of the earth and collide with other plates to form different types of plate margins.
-3 marks

13.Discuss shield volcano, its size, shape and composition with neat diagram.

Figure and Definition (1 mark)-A shield volcano is a broad, gently sloping volcanic mountain formed by low-viscosity basaltic magma that spreads widely.Shape and Slope (2 marks). Built by fluid basaltic lava flowing easily over large areas. Slopes are gentle (6º–12º) from the horizontal-Lava erupts from vents or fissures, forming craters at the summit. Formed by repeated lava flows. Size (1 mark)-Can be enormous in size — e.g., Mauna Kea (Hawaii) rivals Mount Everest in height when measured from its base on the seafloor. Common in Hawaii and Iceland. Composition (3 marks)- Formed from basaltic (mafic) magma with ~50%

silica, low viscosity, and high magnesium and iron content. Produces dark-colored basalt rock- Basaltic lava flows easily and erupts gently from fissures.Style of Eruption (1 mark)- Gentle, non-explosive eruptions; lava flows move slowly, causing limited damage and allowing evacuation.

14. Discuss the differences between tectonic and volcanic earthquakes. Describe their causes, typical locations, and key characteristics.

Definition & Main Distinction (1 mark)- Tectonic earthquakes: Caused by brittle rupture/slip along faults from accumulated elastic stress due to plate motion.Volcanic (magmatic) earthquakes: Caused by movement or pressure of magma/gas beneath the surface.2. Magnitude & Intensity (1 mark)- Tectonic: Produce the world's strongest earthquakes.Volcanic: Generally smaller, often occur in swarms before eruptions.3. Typical Locations (1 mark)-Tectonic: Found mainly along plate boundaries (convergent, divergent, transform).-Volcanic: Occur in volcanic regions—hot spots, subduction zones, and spreading centers.4. Causes & Mechanism – Tectonic Earthquakes (2 marks.-Continuous plate movement builds stress along locked faults.When stress > friction, rocks fracture and slip, releasing energy as seismic waves (elastic rebound).Common atConvergent zones (Benioff zone) – strong quakes.Transform faults – horizontal slip (e.g., San Andreas).Divergent ridges – shallow, small quakes.5. Causes & Mechanism – Volcanic Earthquakes (2 marks)- Generated by magma movement and pressure changes within the crust.Often precede volcanic eruptions (earthquake swarms signal magma ascent).Common at subduction zones, hot spots, and divergent boundaries.6. Summary/Conclusion (1 mark)- Tectonic earthquakes result from mechanical stress and fault rupture, while. Volcanic earthquakes signal magma movement within or near volcanoes.Both reflect the Earth's dynamic internal processes.

Part C
Essay in 400 words
Answer any 1 question
Each question carries 12 marks

15. Explain the concept of plate tectonics, and discuss the landforms associated with various plate margins.

Concept of Plate Tectonics (4 marks)-The theory of plate tectonics explains Earth's dynamic activities — earthquakes, volcanoes, mountain building, and continental drift.The lithosphere (crust + upper mantle, ~75–125 km thick) is broken into tectonic plates. These plates float on the asthenosphere, a weak, partially molten layer (~350 km deep).Plates move slowly (1–16 cm/year), and their interactions at boundaries cause most major geological events.(1 mark for definition, 1 for structure/layers, 1 for movement, 1 for significance).

Plate Boundaries and Landforms – 8 marks

A. Divergent Boundaries (Spreading Centers) – 3 marks Plates move apart, magma rises to form new lithosphere. Ocean–Ocean: Mid-ocean ridges and rift valleys; sea-floor spreading, shallow earthquakes, volcanoes.Continent–Continent: Rift valleys (e.g., East African Rift), earthquakes, rising magma, volcanic activity.(marks: 1 for process, 1 for oceanic example, 1 for continental example)

B. Convergent Boundaries – 3 marks, Plates move toward each other; one may subduct beneath the other, forming trenches and mountains.Ocean–Ocean: Island arcs and

trenches; subduction, deep earthquakes, volcanoes.Ocean–Continent: Coastal mountains (e.g., Andes); subduction, volcanoes, deformation. Continent–Continent: Collision forms large mountain ranges (e.g., Himalayas), crustal thickening, folding, thrust faulting(marks Division: 1 for process, 1 for subduction examples, 1 for continental collision)
C. Transform Boundaries – 2 marks-Plates slide horizontally past one another along strike-slip faults. Ocean–Ocean: Offset mid-ocean ridges, earthquakes.Continent–Continent: Fault zones (e.g., San Andreas Fault); earthquakes, deformation, uplift.(marks division: 1 for process, 1 for examples/effects)

16. Critically assess the overall contributions and limitations of different earthquake measurement techniques in studying earthquakes.

Introduction – Importance of Measurement (1 mark)

Earthquake measurement techniques form the foundation of seismology, enabling scientists to record, quantify, and analyze seismic energy and Earth's structure, though accurate short-term prediction remains limited.

2**. Measurement Scales - types-– Contributions & Limitations (4 marks)**

Scale/Technique	Contribution	Limitation
Intensity Scales	Measure strength based on damage observed.	Subjective – depends on distance, ground type, and building quality.
Richter Scale (1935)	Provides quantitative magnitude using wave amplitude.	Doesn't represent total energy released; short quakes may appear stronger.
Moment Magnitude Scale	Most accurate; based on fault slip and rupture area – measures total energy.	Requires complex data collection; harder to calculate quickly.

Both Richter and Moment Magnitude scales show a 30× energy increase per unit rise in magnitude.
Largest recorded quakes: 8.5–8.7 magnitude. *(mark division: 1 for overview, 1 for each scale pair: intensity, Richter, moment magnitude)*

3. **Seismic Wave Recording & Earth Structure (3 marks)**

1. Locating the Epicenter- Seismographs record P and S waves on a seismogram.The time gap between P and S wave arrivals gives the distance to the epicenter.Data from three or more stations locate the quake precisely.
2. Mapping Earth's Interior

Seismic waves refract/reflect at layer boundaries.S waves not passing through liquids revealed a molten outer core.Helped discover key boundaries like the Moho (crust–mantle boundary).*(mark division : 1 for method, 1 for epicenter location, 1 for internal structure mapping)*

4**. Earthquake Prediction – Critical Assessment (4 marks)**

A. Long-Term Prediction (2 marks). Based on recurrence history along active faults (e.g., San Andreas).Identifies high-risk zones and informs building codes.Limitation: Gives only probabilities, not specific times or locations.
B. Short-Term Prediction (2 marks). Monitors precursor signals such as:Foreshocks, crustal distortion, radon gas, water-level changes, and animal behavior.Limitation: Unreliable – precursors are inconsistent.E.g., Haicheng (1975) successfully predicted, but Tangshan (1976) failed, causing 250,000+ deaths.*(Mark division: 2 for long-term, 2 for short-term with examples)*